

Proteomic profiles and leaf antioxidative metabolism in cacao juvenile plants subjected to Pb and Pb+Fe doses and to soil flooding

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ABSTRACT

Under natural conditions, plants can be affected by several types of abiotic stresses, such as flooding and the accumulation of toxic metals in the soil. Pb is a non-essential metallic element for plants and is highly toxic for living beings. Fe, as an essential micronutrient, is involved in fundamental plant processes, such as photosynthesis. The main objective of this work was to evaluate the proteomic profiles and the activity of key enzymes of leaf antioxidant metabolism in juvenile plants of the CCN 51 cacao genotype, grown in flooding soil and subjected to a dose of Pb and to different equimolar doses of Pb+Fe. Doses of Pb and Pb+Fe did not interfere with the predawn leaf water potential (Ψ_{WL}) of plants, but soil flooding was a key factor for the decline of Ψ_{WL} . There was a differential accumulation of Pb and Fe in the roots and leaves, both in flooded soil and non-flooded. Most of Fe and Pb uptake by the root system predominantly accumulated in the roots and only a small fraction was translocated to the shoot, regardless of the water content in the soil. Flooding and the variation of Pb+Fe doses in the soil did not change the transport of Pb to the plant shoot since there were no major alterations in the levels of Pb in the leaves. Leaf enzymatic antioxidant metabolism was altered, mainly by flooding, in relation to Pb or Pb+Fe doses applied in soil. The time of exposure of plants to stressor factors was important for the changes in the activity of key enzymes of the leaf antioxidant metabolism. Soil flooding, with doses of 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil, at 6 and 24 h after treatment application, had a greater influence on Ψ_{WL} , antioxidant enzymatic metabolism and leaf proteome, and affected proteins related to photosynthesis, DNA, antioxidative system and carbohydrate metabolism, whose abundance was overexpressed or repressed compared to the control.

1. Introduction

Under field conditions, plants may be subjected to multiple stresses, such as metal toxicity and water stress, caused by flooding and/or soil water deficiency. Flooding is common in hydromorphic soils, in regions with high rainfall, or in areas with excessive irrigation (Pezeshki, 2001). One of the main effects of flooding is O₂ deprivation in the root zone (anoxia) (Folzer et al., 2005).

Flooding is one of the crucial stressors that has an appreciable effect on plant growth, leading to decreased yield and production (Normile, 2008). Globally, around 13% of total land and nearly 10% of agricultural land is adversely affected by flooding (Cramer et al., 2011). *Theobroma cacao* (cacao) generally grows in regions with annual rainfall of

1500 and 2000 mm, being considered a species intolerant to flooding and should be cultivated in well-drained soils (Sena Gomes and Kozłowski, 1986). Cacao grown in shallow soils and/or with drainage problems are periodically flooded in occasions of high precipitation, creating an anoxic condition in the soil, depriving the root system of oxygen, and inducing it to anaerobic respiration, which compromises growth and plant development (Bertolde et al., 2014).

Several morphophysiological characteristics of the plant are affected by flooding (Kozłowski, 2002; Jia et al., 2021). Physiological responses to root zone hypoxia or anoxia include (i) decline in photosynthesis, attributed primarily to decreased CO₂ uptake as a result of stomatal closure; (ii) changes in source-sink relationships; (iii) decrease in nutrient uptake; and (iv) changes in hormonal balance, with increased

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synthesis of ethylene and abscisic acid (ABA), among others (Pezeshki, 1993; Kozłowski, 1997; Voesenek and Bailey-Serres, 2013). In cacao, some of the main plant responses to this type of stress include (i) partial or total closure of stomata, followed by a decrease in photosynthesis; (ii) changes in the source-sink relationship; (iii) decrease in nutrient uptake; (iv) development of hypertrophied aerenchyma and lenticels (v) and changes in biomass production (Bertolde et al., 2012; Da Silva Branco et al., 2017).

Iron (Fe) is an abundant metallic element, estimated at 3.8% of the average weight of soils, but it is predominantly present in insoluble oxidized forms, which are mostly unavailable for plant uptake (Sanchez-Sanchez et al., 2002). The productivity and sustainability of cacao in tropical soils are affected by Fe contents (Baligar et al., 2015). Cacao is grown on highly weathered Latosols and Argisols that are micro-nutrient deficient (Fageria and Baligar, 2008). Fe deficiency is widely reported in highly weathered tropical acid soils where cacao is grown (Hartemink, 2005). In Bahia, Brazil, the availability of Fe in soils, where cacao is grown, ranges from 36 to 230 mg kg⁻¹ soil (Souza Junior et al., 1999). Fe occurs in multiple forms in soils and the dominant form is pH dependent (Lindsay and Schwab, 1982). Soil pH, redox potential and soil organic matter content profoundly affect Fe bioavailability (Fageria et al., 2002). Plants uptake Fe only in the Fe²⁺ form (Kliman, 1937) and its availability decreases 1000 times for each pH unit increase between 4 and 9 (Lindsay, 1979). Fe deficiency occurs not because of its scarcity in the soil, but due to various soil and plant factors that affect its availability (Fageria et al., 2002).

In acidic soils, hypoxia/anoxia, and the pH reduction, resulting from flooding, can lead to the reduction and solubilization of large amounts of Fe (Ponnamperuma, 1972). Fe, being an essential nutrient for plants, is involved in fundamental processes such as (i) photosynthesis, where it alternates as Fe²⁺ and Fe³⁺, by reversible redox reaction due to electron transfer; (ii) chlorophyll and several proteins biosynthesis; (iii) respiration; (iv) N metabolism; (v) syntheses of DNA and hormones and (vi) stages of metabolism of photosynthetic pigments (Briat et al., 2007). Thus, Fe is a fundamental micronutrient for plant growth and development.

Lead (Pb) interferes with several physiological attributes (Saleem et al., 2019); however, consolidated information on plant-Pb interactions related to Pb translocation in tissues, along with its impact on the antioxidant system, DNA damage and metabolism of proteins are still quite limited (Kumar et al., 2018). Its toxicity is dependent on exposure time and metal content. Exposure of plants to high contents of Pb disturbs water and nutritional relationships and causes oxidative damage. The decrease in plant growth under Pb stress is mainly due to the interference of Pb in enzymatic activities, in the permeability of cell membranes and in stomatal closure, by induction of ABA, and by the negative relation of Pb and K in plants (Zulfiqar et al., 2019).

Pb and its interactions with plants lead to the production of reactive oxygen species (ROS), whose excess, not eliminated by the enzymatic and non-enzymatic antioxidative system, alters the cellular ultrastructure, damaging cellular organelles and biomolecules (Yadav, 2010). These harmful effects can result in the inhibition of normal cellular functioning, due to physiological and molecular changes, and compromise the overall performance of the plant (Kumar and Prasad, 2018). The antioxidant system, through the action of key enzymes such as superoxide dismutase (SOD), catalase (CAT), ascorbate peroxidase (APX), and guaiacol peroxidase (GPX), among others, eliminates excess ROS and protects the plant against damage oxidative agents (Gill and Tuteja, 2010). The activity of these enzymes maintains the integrity of cell membranes and important molecules such as DNA and proteins, preventing lipid peroxidation and cell death (Gill and Tuteja, 2010) and therefore reducing the damage caused by Pb.

Proteomics analysis provides a powerful tool for analyzing molecular mechanisms of plant response against stresses, as well as being a way to link gene expression with cellular metabolism in the post-genomic era of rapid processing (Ahsan et al., 2009; Liang et al., 2013). The integration

of genomic data with proteomic profiles can provide a better understanding of the molecular mechanisms of responses to flooding, associated with Pb and Pb+Fe doses applied to the soil, in cacao plants. The amount of protein is not always related to the amount of mRNA, especially for low-abundance proteins (Fernie and Stitt, 2012). Furthermore, many proteins undergo post-translational modifications, such as removal of signal peptides, phosphorylation, and glycosylation, which are extremely important for protein activities and subcellular localization (Fernie and Stitt, 2012).

Theobroma cacao beans (fermented and dried seeds) have a high economic value due to butter extraction and chocolate production, although butter is not the only commercial product extracted from its beans, being used also in the cosmetics, medicine, and food industries such as jellies, beverages, and ice cream manufacture (Almeida and Valle, 2007). In recent years, there has been concern about food safety in relation to the presence of toxic metallic elements in cacao beans, mainly with Pb and Cd contents (Bertoldi et al., 2016; Arévalo-Gardini et al., 2017). These heavy metals are extracted from the soil via root system, in areas whose sources of contamination are the rock of origin, during soil formation, and anthropic actions in the crop, mainly the applications of fertilizers and pesticides (Gramlich et al., 2017; Apraiz Muñoz et al., 2021). The levels of these toxic metals in cacao beans, from different origins around the world, have exceeded the recommended limits, causing concern for human health (Bertoldi et al., 2016). In addition, the levels of Pb and Cd found in beans from cacao producing countries in Latin America are higher compared to other cacao producing countries in the world (Bertoldi et al., 2016). This, in turn, can compromise their exports, mainly to the European market (Arévalo-Gardini et al., 2017).

Both Pb and Fe are uptake by the root system of plants as divalent ions. Under flooding conditions, Fe is reduced from Fe³⁺ to Fe²⁺, thus increasing its availability to plants. Pb, as it has no specific transporter in plants, competes for divalent ion transporters. Therefore, Fe, in the form of Fe²⁺, can compete with Pb²⁺ uptake and mitigate Pb toxicity in plants. Nonetheless, a study is needed to elucidate whether there is a mitigation of Pb toxicity by Fe in cacao, in order to reduce the risks to human health due to the accumulation of Pb in beans and consequent contamination of its derivative products. Given this context, the present work hypothesizes that Pb²⁺, when competing with Fe²⁺, during its uptake via root system, in conditions of sporadic flooding, in which the availability of Fe²⁺ in the soil solution may favor the uptake of Fe²⁺ rather than Pb²⁺, will mitigate Pb toxicity in juvenile plants of the cacao genotype CCN 51.

The main objective of this work was to evaluate the proteomic profiles and the activity of key enzymes of leaf antioxidant metabolism in juvenile plants of the CCN 51 cacao genotype, grown in flooded soil and subjected to a dose of Pb and different doses of Pb+Fe.

2. Material and methods

2.1. Grown conditions and plant material collection

The experiment was conducted in a greenhouse at the State University of Santa Cruz (UESC), Ilhéus, Bahia, Brazil. Seeds of the cacao genotype CCN 51, derived from controlled pollination, were used. The genotype CCN 51 has desirable agronomic characteristics because it is self-compatible, highly productive, tolerant to *Moniliophthora perniciosa* and has a butter content in its beans. Therefore, this genotype is widely cultivated by cacao producers.

The seeds, pre-germinated in sawdust moistened with water, were transferred to plastic pots, without perforations for water drainage, with a capacity of 14 kg, containing soil classified as textural loam-clay-sandy (Supplementary Table 1), with acidity and pH previously corrected and fertilized, according to the nutritional requirements of the cacao tree (Souza Júnior et al., 2011). Before planting, the soil was fertilized with P, K, B and Mo (0.96, 134, 3.4 and 0.18 mg dm⁻³ soil, respectively) and a

mixture of CaCO_3 and MgCO_3 , important to reach the $\text{Ca}^{2+}:\text{Mg}^{2+}$ ratio of 4:1 and raising the base soil saturation value to 80%, which results in an increase in pH (Supplementary Table 1). In the cover fertilization, the soil was fertilized with N, K and S (67, 54 and 18 mg dm^{-3} soil, respectively), 15 days after transplanting and every 15 days.

At 120 days after transplanting, 1 L of PbCl_2 solution and/or 0.5 L of FeCl_2 solution were added per pot, as sources of Pb and Fe, respectively, according to the equimolar treatments of Pb or Pb+Fe, always keeping the Pb dose constant (2 mmol Pb kg^{-1} soil), except in the control. Immediately afterwards, half of the plants were maintained with soil moisture near field capacity (non-flooded) and the other half was subjected to flooding, making a total of 12 treatments (T1 - Control treatment - soil moisture maintained near field capacity (FC) and without addition of Pb or Pb+Fe in soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in soil; T3 - Treatment with soil moisture maintained near FC and with 2 mmol Pb kg^{-1} soil; T4 - Flooded treatment with 2 mmol Pb kg^{-1} soil; T5 - Treatment with soil moisture maintained near FC and with 2 mmol Pb kg^{-1} soil + 0.5 mmol Fe kg^{-1} soil; T6 - Flooded treatment with 2 mmol Pb kg^{-1} soil + 0.5 mmol Fe kg^{-1} soil; T7 - Treatment with soil moisture maintained near FC and with 2 mmol Pb kg^{-1} soil + 1 mmol Fe kg^{-1} soil; T8 - Flooded treatment with 2 mmol Pb kg^{-1} soil + 1 mmol Fe kg^{-1} soil; T9 - Treatment with soil moisture maintained near FC and with 2 mmol Pb kg^{-1} soil + 1.5 mmol Fe kg^{-1} soil; T10 - Flooded treatment with 2 mmol Pb kg^{-1} soil + 1.5 mmol Fe kg^{-1} soil; T11 - Treatment with soil moisture maintained near FC and with 2 mmol Pb kg^{-1} soil + 2 mmol Fe kg^{-1} soil; and T12 - Flooded treatment with 2 mmol Pb kg^{-1} soil + 2 mmol Fe kg^{-1} soil), with 20 plants per treatment, totaling 240 plants. The experiment lasted 15 days after application of treatments (AAT). In pots with flooding, the water level applied was maintained at about 2 cm above the soil surface. During the entire experimental period, the plants were irrigated with rainwater.

Photosynthetically active radiation (PAR), temperature (T) and air relative humidity (RH) inside the greenhouse were recorded daily throughout the experimental period. The minimum and maximum PAR values were 2.7 and 8.3 $\text{mol photons m}^{-2} \text{ day}^{-1}$, respectively, with a daily average value of $5.9 \pm 0.4 \text{ mol photons m}^{-2} \text{ day}^{-1}$. The minimum and maximum T values were 24.8 °C and 30.6 °C, respectively, with a daily mean value of 27.4 ± 0.3 °C. The minimum and maximum RH values were 65.2% and 91.2%, respectively, with a daily mean value of $76.2 \pm 1.4\%$.

Mature leaves (2nd or 3rd leaf from the apex of the plants) of all treatments were collected at 0, 3, 6, 12, 24 and 48 h AAT for biochemical and molecular analyses. In addition, the whole plant of all treatments was collected at 15 days AAT, for analysis of Pb and Fe contents in roots and leaves. Immediately after the collections, the plant material was wrapped in aluminum foil, immersed in liquid nitrogen, stored in an ultra-freezer at -80 °C and subsequently lyophilized and stored in a freezer at -20 °C.

2.2. Predawn leaf water potential (Ψ_{WL})

Ψ_{WL} measurements were performed at the 1st, 7th, and 14th days AAT, between 2:00 and 4:00, using a model 1000 pressure chamber instrument (PMS Instrument Company, USA). The pressurization was performed slowly, and the time of leaves collection and measurement was as short as possible (Scholander et al., 1965). During the sampling period the minimum air temperature was around 24 °C and the maximum relative humidity was 90%.

2.3. Analysis of Pb and Fe contents

Juvenile cacao plants were collected at 15 days AAT. Soon after, the plants were separated into roots and leaves. The roots were subjected to successive washes in the following order: running water (2x), neutral detergent solution (0.1%, 2x), distilled water (2x), hydrochloric acid (3%, 2x) and distilled water (2x), for the removal of metals that were not

absorbed by the root system. Immediately afterwards, the plant material was stored separately in paper bags, according to the treatments, and then placed in a forced air circulation oven at 70 ± 5 °C until constant mass. Subsequently, the dry plant material was ground in a Willey mill, model MA-340, using 20 mesh sieves. Soon after, the ground plant material was subjected to nitroperchloric digestion (3:1, v.v.). After digestion, the levels of Pb and Fe in the different organs (root and leaf) of the plants, of all treatments, were determined using optical emission spectrometry with inductively coupled plasma (ICP-OES, Varian 710-ES).

2.4. Antioxidative metabolism

For the analysis of the activity of antioxidant metabolism enzymes, samples of lyophilized leaves, collected in all treatments at intervals of 0, 3, 6, 12, 24 and 48 h AAT were used. The activities of superoxide dismutase (SOD, EC 1.15.1.1), ascorbate peroxidase (APX, EC 1.11.1.11), catalase (CAT, EC 1.11.1.6) and guaiacol peroxidase (GPX, EC 1.11.1.7) were determined according to Giannopolitis and Ries (1977), Nakano and Asada (1981), Havir and Mchale (1987) and Kar and Mishra (1976), respectively.

2.5. Principal components multivariate analysis

In order to evaluate the interrelationship between treatments and leaf activities of SOD, APX, CAT and GPX in all treatments, a biplot analysis was made based on principal components multivariate analysis. Initially, the values of each variable were standardized to mean 0 and variance 1 using the equation $Z_{ij} = (X_{ij} - \mu_j)/S_j$, where X_{ij} is the value of the i th observation of the variable X_j , and μ_j and S_j is the mean and the standard deviation of X_j . Then, the standardized values were submitted to multivariate factorial analysis using factors loadings > 0.70 as a criterion. For each variable selected by the factor loading value, the average of each treatment was used as input data in the biplot analysis, and the values were obtained from the decomposition into singular values.

2.6. Proteomic analysis

From the results of the biplot analysis, between the treatments and the leaf activities of SOD, APX, CAT and GPX, two contrasting treatments were selected [T1 (control) - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe in soil and T6 - soil flooded with 2 mmol Pb kg^{-1} soil + 0.5 mmol Fe kg^{-1} soil], to be used in proteomic analysis. The enzymatic activities were evaluated at 6 h and 24 h AAT. To analyze the leaf proteomic profiles of the selected treatments, the protocol developed by Pirovani et al. (2008) was used.

2.6.1. Extraction of total proteins from leaf samples

Initially, 0.2 g of lyophilized biomass of the samples for each treatment was macerated, using liquid nitrogen and polyvinylpyrrolidone (PVPP), with a mortar and pestle, until reaching the texture of a fine powder. After maceration, the samples were stored in 2 mL Eppendorf micro tubes, so that they could be subjected to several washes. Subsequently, the material was washed with 100% acetone and 0.07% β -mercaptoethanol and centrifuged at 14,000 g for 10 min at 4 °C five times. Between one wash and another, the samples were centrifuged under the same initial conditions and the supernatant was discarded. Then, the pellet was dried at room temperature for 1 h and then washed with 10% TCA in acetone containing 0.07% β -mercaptoethanol. The washes were repeated four times until the leaf samples lost their color. Between one wash and another, the samples were centrifuged, under the same conditions as in the previous step, and homogenized using a vortex. Soon after, washings were performed with 10% TCA in water, followed by homogenization by sonication (8 pulses of 5 s, amplitude of 70%, with intervals of 10 s in the ultraprocessor (Gex 130, 130 W) and

subsequent centrifugation to discard the supernatant. This step was repeated twice, and at the end, a first precipitation was performed, for approximately 1 h at 20 °C, using the same centrifugation conditions. Then, the supernatant was discarded, and the samples were washed twice with 80% acetone. After each wash, the precipitate was resuspended using the vortex and sonicated when necessary. In the last wash, the supernatant was discarded, and the pellet dried at room temperature. Subsequently, the pellet was resuspended in 1 mL of dense SDS extraction buffer (30% sucrose, 2% SDS, Tris-HCl (0.1 M and pH 8.0) and 5% β -mercaptoethanol). Then, the sample was submitted to sonication on ice (4 pulses of 10 s, amplitude of 70%, with intervals of 10 s), followed by the addition of 1 mL of saturated phenol, pH 8.0.

Afterwards, the sample was centrifuged at 10,000 g at 4 °C for 15 min. The supernatant was then transferred to a new tube and stored on ice. The phenol phases were pooled and incubated at -20 °C overnight with five volumes of 0.1 M ammonium acetate in methanol. Protein precipitates were recovered by centrifugation at 14,000 g at 4 °C for 10 min. Then, the proteins were washed twice with 0.1 M ammonium acetate in methanol and twice with 80% acetone. During each wash cycle, the sample was centrifuged at 14,000 g at 4 °C for 5 min. Finally, the pellets were dried at room temperature. Soon after, the pellet was resuspended in 400 μ L of 8 M urea. At the end, the proteins from the leaf samples were stored in a freezer at -20 °C until the moment of use.

2.6.2. Quantification of total proteins and electrophoresis in SDS-PAGE

The obtained protein extracts were quantified with the 2D Quant Kit (GE Healthcare Life Sciences), following the manufacturer's recommendations. The standard used was different concentrations of bovine serum albumin (BSA), which served as the basis for generating a standard dosage curve for the quantification of leaf samples. Both the curve and the samples were prepared in triplicate, using 2 mL Eppendorf micro tubes. Initially, 100 μ L of precipitating solution was added to all micro tubes (standard and samples). Soon after, 2 μ L of samples resuspended in rehydration buffer and the corresponding concentrations of the standard were added to the micro tubes of the curve, being briefly homogenized in vortex. Subsequently, 100 μ L of co-precipitant solution were added to each microtube, followed by inversion homogenization. Then, the micro tubes were subjected to centrifugation at 4 °C for 15 min at 14,000 g, followed by complete disposal of the supernatant. Soon after, 20 μ L of copper solution and 80 μ L of distilled and autoclaved water were added to the pellet, followed by vortex homogenization to dissolve the precipitate. Shortly thereafter, 200 μ L of the "working color reagent" (reagent A + reagent B) were added to all Eppendorf's and the solution formed was incubated for 15 to 20 min at room temperature. Then, the solutions were transferred to ELISA microplates and read in a spectrophotometer (480 nm), generating a standard curve.

2.6.3. SDS-PAGE gel electrophoresis

After the quantification step, the samples were analyzed by SDS-PAGE (Sodium Dodecyl Sulfate - Polyacrylamide Gel Electrophoresis) in mini electrophoresis vats (Omniphor), with 8 $\times$ 10 cm gels, containing 12.5% sodium acrylamide according to Laemmli (1970), where it was stained with 0.08% colloidal Coomassie according to Neuhoff et al. (1988). Thirty μ g of each sample were used and, from this gel, it was possible to observe the profile of total protein bands.

2.6.4. Mass spectrometry (MS)

The solution containing the digested peptides was desalted using tips with C18 resin (10 μ L; Millipore® Ziptips C18). Peptides were eluted in 50 μ L of a solution containing 75% acetonitrile, 25% water and 0.1% formic acid. The peptides were then submitted to analysis on a liquid chromatography system (Agilent 1290 Infinity II HPLC) coupled to a quadrupole/Time-of-Flight mass spectrometer (Agilent 6545 LC/QTOF). Samples were separated using a reversed phase column (C18; AdvanceBio Peptide Mapping 2.1 $\times$ 250 mm; Agilent), maintaining a temperature of 55 °C. A 20 min gradient was applied with mobile phases

A (H₂O and 0.1% formic acid) and B (acetonitrile and 0.1% formic acid). The B-phase percentages along the gradient were 5% to 35% (1–10 min.), 35% to 70% (11–14 min.), 70% to 100% (16 to 18 min.) and 100% (16–20 min.). In addition, a final period of 5 min was programmed for column stabilization. Then, the samples were injected in 3 technical replicates.

The samples were injected into the QTOF through an electrospray source, using the AutoMSMS acquisition mode, with a maximum selection of 10 precursors per cycle. The parameters for selection of precursors were threshold of 1000, 10,000 counts/spectrum, purity stringency of 100%, purity cutoff of 30%, peptide isotope model, charge preference of 2, 3, >3 and unknown. The collision energy (in V) was configured to be applied according to the formula: $(slope) * (m/z)/100 + Offset$, where m/z represents the mass/charge ratio of the precursor and the slope and Offset vary from 3.1 to 5 and from -4, 8 to 10, respectively, depending on whether the precursor charge is 2, 3, >3, or unknown. The instrument parameters used were gas temperature of 325 °C, gas flow of 13 L/min, capillary voltage of 4000 V and skimmer voltage of 56 V. Nitrogen gas was used for collision induced dissociation. Instrument control (HPLC and QTOF) and parameter configuration were performed using the Agilent MassHunter Acquisition software.

2.6.5. Identification of peptides in protein bank

The resulting spectra were processed in triplicate for peptide identification, using Spectrum Mill software (Rev B.06.00.203 SP1; Agilent). Parameters for spectra extraction were MSNoiseThreshold of 10 counts, carbamidomethylation fixed modifications, precursor MH+ from 200 to 6000, retention time tolerance of $\pm$ 60 s, m/z tolerance $\pm$ 1.4 and load forerunner find. After extracting the MS/MS spectra, a database search was performed. The database used was *T. cacao* downloaded from UniProt (<https://www.uniprot.org>). Parameters for comparing MS/MS spectra in the protein bank were maximum number of cleavages missed: 4, fixed modifications: carbamidomethylation (C), variable modifications: oxidized methionine (M), pyroglutamic acid (N-termQ), deamidated (N), phosphorylated S (S), phosphorylated T (T), phosphorylated Y (Y), combined minimum peak intensity: 10% and precursor mass tolerance: $\pm$ 10 ppm. Search results were validated and filtered for those that had a false positive rate (FDR) of less than 1%. Results were exported in protein-protein comparison mode in MPP APR file format.

2.6.6. Identification of differentially abundant proteins

Statistical analyses to identify differentially abundant proteins were performed using Mass Profiler Professional 15.1 software (MPP; Agilent). Protein abundance was calculated using the median of peptide abundance. One experiment was performed for each treatment. In each experiment, the results of the triplicates of the control with the triplicates of the corresponding treatment were compared. Proteins were filtered by peptide frequency, according to the following criterion: retain proteins that appear in at least 100% of triplicates, in at least one condition. Statistical significance analysis was performed using the unpaired *t*-test (treatment vs. control), *p*-value calculation was asymptotic and multi-test correction was performed using the Benjamini-Hochberg method. Only proteins that had a *p*-value < 0.05 and fold change $\geq$ 1.5 were considered differentially expressed. Principal component analysis, Venn diagram and heat map were automatically generated in the same software. For the Venn diagram and principal component analysis, all identified proteins were considered. For the heat map, only those proteins that statistically satisfied the criteria of *p*-value < 0.05 and fold change $\geq$ 1.5 were considered. The parameters used were normalized intensity values, Euclidean metric distance and Ward linkage method.

2.7. Statistical analysis

The experimental design was in randomized blocks, in a factorial scheme formed by two water regimes [flooded soil and non-flooded soil (soil moisture close to field capacity)] and six doses [1 (Pb) + 4 (Pb+Fe)

+ control], making a total of 12 treatments, 20 repetitions and one plant per experimental unit. The results were submitted to analysis of variance (ANOVA) and the averages, when relevant, compared by the Scott-Knott test (comparison of water regimes and doses of Pb and Pb+Fe).

3. Results

3.1. Predawn leaf water potential (Ψ_{WL})

There were no significant differences ($p < 0.05$) in Ψ_{WL} for treatments T1, T3, T5, T7, T9 and T11 at 1-, 7- and 14-days AAT (Table 1). For treatments T2, T4, T6, T8, T10 and T12, there was a significant reduction ($p < 0.05$) in the Ψ_{WL} values with AAT days, with the lowest Ψ_{WL} value observed at 14 days AAT (Table 1). The lowest Ψ_{WL} value was found for treatment T6 followed by T4 (Table 1). On the first AAT day, there were no significant differences ($p < 0.05$) for Ψ_{WL} between treatments compared to the control. At 7 days AAT the lowest value of Ψ_{WL} was observed for the T12 treatment. On the other hand, at 14 days AAT, the lowest value of Ψ_{WL} was verified for T6. Lower Ψ_{WL} values were found for the flooded treatments than the treatments with soil moisture maintained near FC at 7- and 14-days AAT (Table 1).

3.2. Pb and Fe contents

In plants subjected to flooding, Pb and Fe contents, both in leaves and roots, differed significantly ($p < 0.05$) among treatments at 15 days AAT. Fe content in roots was higher than in leaves (Table 2) on a 100x scale. The highest Fe content in leaves was observed for the T6 treatment (Table 2), while the highest Fe content in roots was verified for the T3, T8 and T11 (Table 2). In leaves, the lowest Fe content was observed for T11 followed by treatment T10 (Table 2), while in roots the lowest Fe content was observed in treatment T2 (Table 2). On the other hand, in relation to Pb, a higher Pb content was also observed in the roots (Table 2), when compared to the leaves (Table 2). In the leaves, no significant differences were observed ($p < 0.05$), for most of the

Table 1

Predawn leaf water potential (Ψ_{WL}) in juvenile plants of CCN 51 cacao genotype subjected to flooding and doses of Pb or Pb+Fe in the soil at 1-, 7- and 14-days after application of treatments (AAT). Mean values of three replications ($\pm$ SE). Uppercase letters indicate comparisons between water regimes within each Pb or Pb+Fe dose and lowercase letters comparisons among Pb or Pb+Fe doses within each water regime by the Scott Knott test ($p < 0.05$).

Treatment	Ψ_{WL} (MPa)		
	Time (day)		
	1	7	14
T1(FC)	-0.25 $\pm$ 0.03 ^{Aa}	-0.26 $\pm$ 0.03 ^{Aa}	-0.17 $\pm$ 0.01 ^{Aa}
T2 (Fl)	-0.24 $\pm$ 0.01 ^{Aa}	-0.69 $\pm$ 0.06 ^{Ba}	-0.90 $\pm$ 0.02 ^{Ba}
T3 (FC + 2 Pb)	-0.21 $\pm$ 0.03 ^{Aa}	-0.26 $\pm$ 0.05 ^{Aa}	-0.24 $\pm$ 0.01 ^{Aa}
T4 (Fl + 2 Pb)	-0.22 $\pm$ 0.01 ^{Aa}	-0.92 $\pm$ 0.13 ^{Ba}	-1.30 $\pm$ 0.12 ^{Bb}
T5 [FC + (2 Pb + 0.5 Fe)]	-0.23 $\pm$ 0.01 ^{Aa}	-0.24 $\pm$ 0.04 ^{Aa}	-0.23 $\pm$ 0.01 ^{Aa}
T6 [Fl + (2 Pb + 0.5 Fe)]	-0.24 $\pm$ 0.03 ^{Aa}	-0.84 $\pm$ 0.09 ^{Ba}	-1.38 $\pm$ 0.21 ^{Bb}
T7 [FC + (2 Pb + 1 Fe)]	-0.28 $\pm$ 0.02 ^{Aa}	-0.24 $\pm$ 0.03 ^{Aa}	-0.24 $\pm$ 0.02 ^{Aa}
T8 [Fl + (2 Pb + 1 Fe)]	-0.28 $\pm$ 0.01 ^{Aa}	-0.75 $\pm$ 0.11 ^{Ba}	-0.88 $\pm$ 0.10 ^{Ba}
T9 [FC + (2 Pb + 1.5 Fe)]	-0.26 $\pm$ 0.04 ^{Aa}	-0.23 $\pm$ 0.03 ^{Aa}	-0.23 $\pm$ 0.01 ^{Aa}
T10 [Fl + (2 Pb + 1.5 Fe)]	-0.24 $\pm$ 0.02 ^{Aa}	-0.73 $\pm$ 0.05 ^{Ba}	-1.08 $\pm$ 0.09 ^{Bb}
T11 [FC + (2 Pb + 2 Fe)]	-0.19 $\pm$ 0.05 ^{Aa}	-0.31 $\pm$ 0.04 ^{Aa}	-0.23 $\pm$ 0.01 ^{Aa}
T12 [Fl + (2 Pb + 2 Fe)]	-0.26 $\pm$ 0.02 ^{Aa}	-0.97 $\pm$ 0.14 ^{Ba}	-1.0 $\pm$ 0.14 ^{Ba}

NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe in soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in the soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

Table 2

Fe and Pb contents in leaves and roots of juvenile plants of CCN 51 cacao genotype subjected to flooding and Pb and Fe doses in the soil at 15 days after application of treatments (AAT). Mean values of three replications ($\pm$ SE). Uppercase letters indicate comparisons between water regimes within each Pb or Pb+Fe dose and lowercase letters comparisons between Pb or Pb+Fe doses within each water regime by the Scott Knott test ($p < 0.05$).

Treatment	Fe (mg kg ⁻¹ DW)		Pb (mg kg ⁻¹ DW)	
	Root	Leaf	Root	Leaf
T1(FC)	2433 $\pm$ 671 ^{Aa}	80,0 $\pm$ 10 ^{Aa}	60,0 $\pm$ 7 ^{Ab}	8.05 $\pm$ 2 ^{Aa}
T2 (Fl)	1394 $\pm$ 176 ^{Bb}	76,0 $\pm$ 8 ^{Ab}	73,2 $\pm$ 9 ^{Aa}	4.27 $\pm$ 2 ^{Ab}
T3 (FC + 2 Pb)	2861 $\pm$ 234 ^{Aa}	78,4 $\pm$ 7 ^{Aa}	521 $\pm$ 101 ^{Aa}	7.94 $\pm$ 2 ^{Aa}
T4 (Fl + 2 Pb)	2209 $\pm$ 359 ^{Aa}	69,0 $\pm$ 9 ^{Ab}	741 $\pm$ 66 ^{Aa}	7.94 $\pm$ 2 ^{Aa}
T5 [FC + (2 Pb + 0.5 Fe)]	2347 $\pm$ 186 ^{Aa}	78,3 $\pm$ 9 ^{Bb}	725 $\pm$ 134 ^{Aa}	7.74 $\pm$ 3 ^{Ab}
T6 [Fl + (2 Pb + 0.5 Fe)]	1900 $\pm$ 361 ^{Aa}	175 $\pm$ 6 ^{Aa}	577 $\pm$ 65 ^{Aa}	14.7 $\pm$ 3 ^{Aa}
T7 [FC + (2 Pb + 1 Fe)]	1798 $\pm$ 349 ^{Ba}	82,5 $\pm$ 8 ^{Ab}	770 $\pm$ 57 ^{Aa}	7.41 $\pm$ 3 ^{Aa}
T8 [Fl + (2 Pb + 1 Fe)]	2761 $\pm$ 195 ^{Aa}	86,2 $\pm$ 6 ^{Aa}	695 $\pm$ 87 ^{Aa}	10.1 $\pm$ 2 ^{Aa}
T9 [FC + (2 Pb + 1.5 Fe)]	2163 $\pm$ 170 ^{Aa}	80,4 $\pm$ 9 ^{Aa}	735 $\pm$ 106 ^{Aa}	6.79 $\pm$ 1 ^{Aa}
T10 [Fl + (2 Pb + 1.5 Fe)]	2098 $\pm$ 329 ^{Aa}	51,2 $\pm$ 3 ^{Ab}	671 $\pm$ 91 ^{Aa}	7.20 $\pm$ 2 ^{Aa}
T11 [FC + (2 Pb + 2 Fe)]	2699 $\pm$ 187 ^{Aa}	45,7 $\pm$ 3 ^{Ab}	731 $\pm$ 77 ^{Aa}	4.22 $\pm$ 0.3 ^{Ab}
T12 [Fl + (2 Pb + 2 Fe)]	1929 $\pm$ 403 ^{Aa}	69,6 $\pm$ 4 ^{Aa}	536 $\pm$ 72 ^{Aa}	9.30 $\pm$ 3 ^{Aa}

NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe in soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in the soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

evaluated treatments, in relation to the Pb content (Table 2). As for the roots, there were only significant differences ($p < 0.05$) between treatments T1 and T2, which had the lowest Pb contents (Table 2). Furthermore, in leaves, there was a greater accumulation of Fe than Pb (Table 2). In this case, the Fe content was 10x higher than the Pb content. Furthermore, there was also, in the roots, a greater accumulation of Fe than of Pb (Table 2). In the roots, the Fe content was 3x higher than the Pb content. The addition of Pb+Fe doses in the soil did not influence the increase in Pb content in the roots but interfered with the Fe content (Table 2).

3.3. Antioxidative metabolism enzymes

3.3.1. Superoxide dismutase (SOD) activity

Significant differences ($p < 0.05$) were observed for SOD activity in all treatments and leaves sampling times (Fig. 1). The highest SOD activities were verified for treatments T4 and T6, at 12 h AAT, in relation to the control (Fig. 1D). There was a decrease in SOD activity at 3 h (Fig. 1B) and 6 h AAT (Fig. 1C) and in T3, T5, T7, T9 and T11 with soil moisture near FC at 12 h AAT (Fig. 1D). It was found, at 0 h AAT, that there were few significant differences ($p < 0.05$) among treatments. At this time, there was an increase in SOD activity in treatments T1, T2, T3, T6, T7 and T10 in relation to the control (Fig. 1A). At 3 h AAT, an increase in SOD activity was observed in T1, T4, T6, T7, T8, T9 and T11 compared to the control (Fig. 1B). However, at 6 h AAT, few significant differences ($p < 0.05$) were observed among treatments and control, except for treatments T2, T5 and T8, where there was an increase in SOD

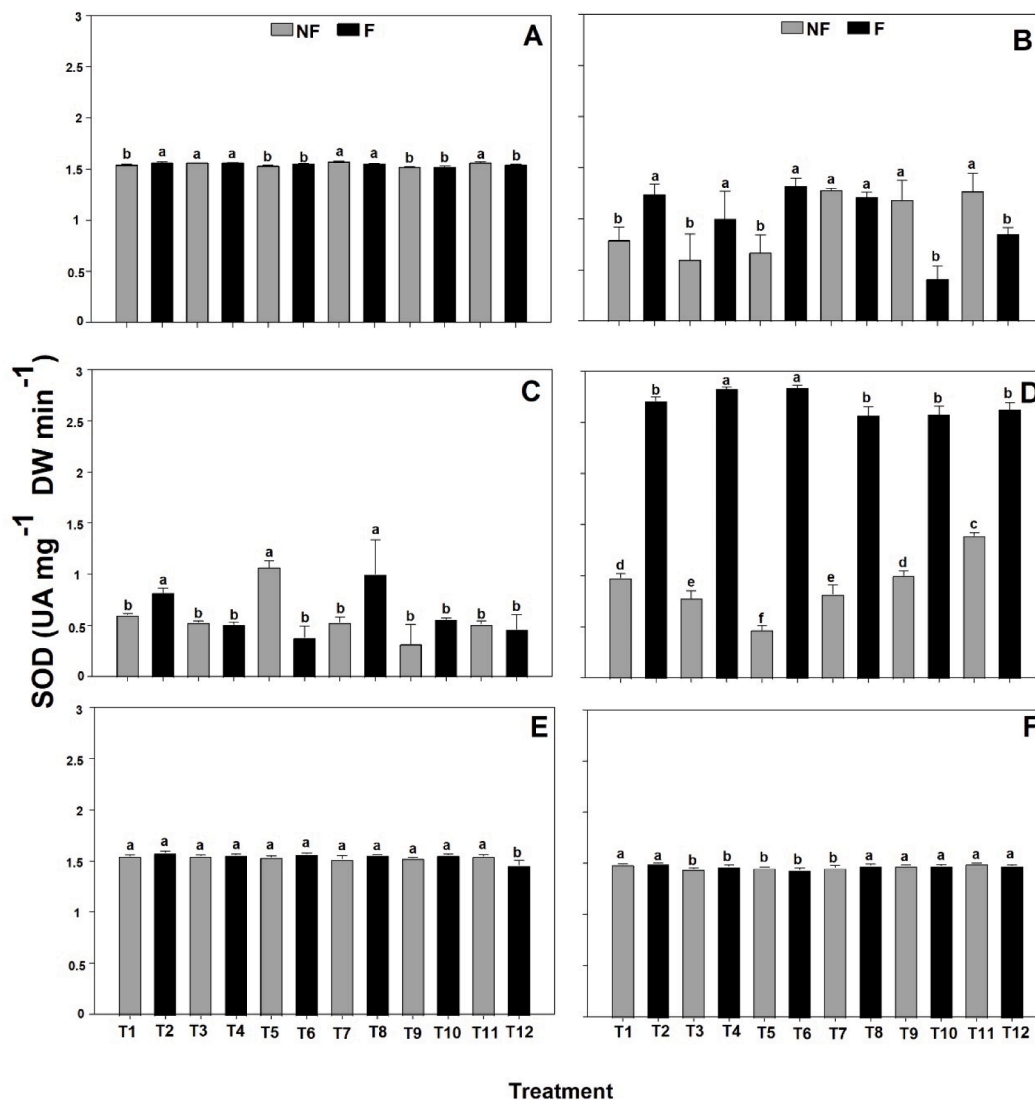


Fig. 1. Superoxide dismutase (SOD) activity in leaves of juvenile plants of CCN 51 cacao genotype subjected to flooding and doses of Pb and Pb+Fe in the soil at 0 (A), 3 (B), 6 (C), 12 (D), 24 (E) and 48 h (F) after application of treatments. Mean values of four replications ($\pm$ SE). Letters indicate comparisons of means among treatments by the Scott-Knott test ($p < 0.05$). NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

activity compared to the control (Fig. 1C). On the other hand, at 12 h AAT, there was an increase in SOD activity for most treatments, except for treatments T3, T5, T7 and T9, compared to the control (Fig. 1D). At 24 h AAT, no significant differences ($p < 0.05$) were observed between treatments in relation to the control, except for T12, where there was a decrease in SOD activity (Fig. 1E). However, at 48 h AAT, few significant differences ($p < 0.05$) were found among treatments and control, except for treatments from T3 to T7, where there was a decrease in SOD activity (Fig. 1F).

3.3.2. Ascorbate peroxidase (APX) activity

Significant differences ($p < 0.05$) were observed for APX leaf activity in all treatments and leaf sampling times (Fig. 2). The highest APX activity was observed for treatments T4 and T6 at 24 h AAT (Fig. 2E), although there was an increase in APX activity at 24 h AAT on all treatments (Fig. 2E). The lowest APX activities were observed at 0 h AAT (Fig. 2A) in which there was a decrease in APX activity in all treatments

in relation to the control. The lowest APX activities were observed in treatments T2, T7, T9, T10, T11 and T12 (Fig. 2A). At 3 h AAT, there were no significant differences ($p < 0.05$) for most treatments compared to the control, except for T3, T7, T8 and T12, where there was an increase in APX activity compared to the control (Fig. 2B). The same fact was observed at 6 and 12 h AAT, where there were no significant differences ($p < 0.05$), for most treatments, in relation to the control, except for treatments T3, T5 and T7 at 6 h AAT (Fig. 2C) and for treatments T8, T10 and T12 at 12 h AAT, where an increase in leaf APX activity was also observed in relation to the control (Fig. 2D). However, at 24 h AAT, there was an increase in APX activity for treatments T2, T3, T4, T6 and T12 and a decrease in APX activity for T11, compared to control (Fig. 2E). However, at 48 h AAT, there were no significant differences ($p < 0.05$) between treatments, except for T5, which showed an increase in APX activity compared to control (Fig. 2F).

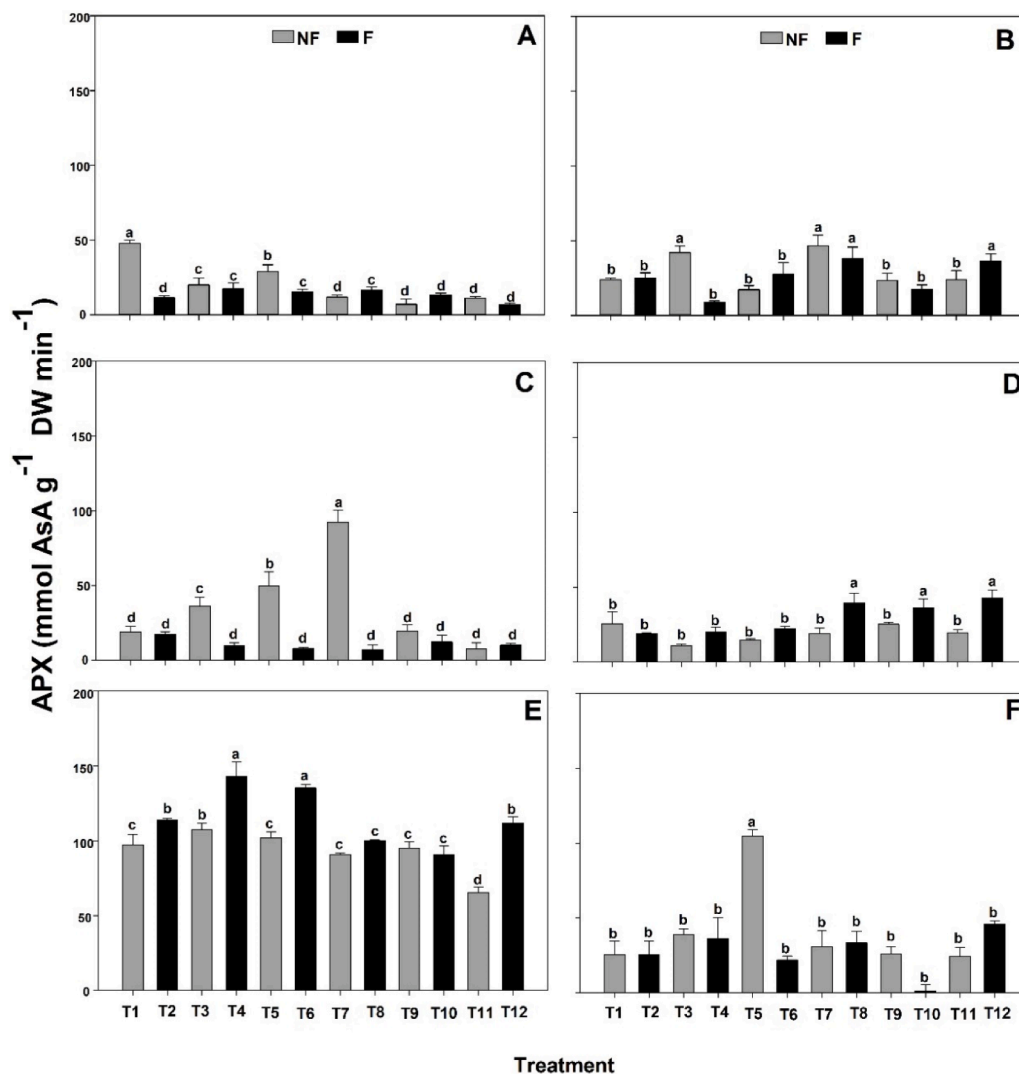


Fig. 2. Ascorbate peroxidase (APX) activity in leaves of juvenile plants of CCN 51 cacao genotype subjected to flooding and doses of Pb and Pb+Fe in the soil at 0 (A), 3 (B), 6 (C), 12 (D), 24 (E) and 48 h (F) after application of treatments. Mean values of four replications ($\pm$ SE). Letters indicate comparisons of means among treatments by the Scott-Knott test ($p < 0.05$). NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

3.3.3. Catalase (CAT) activity

There were significant differences ($p < 0.05$) for leaf CAT activity in all treatments and sampling times (Fig. 3). The highest CAT activity was observed for T6 at 48 h AAT (Fig. 3F). At 0 h AAT, an increase in CAT activity was observed for treatments T2, T6, T8, T9 and T12 in relation to the control (Fig. 3A). At 3 h AAT, an increase in CAT activity was verified for T8, T10 and T12 in relation to the control (Fig. 3B). At this time, there was also an increase in CAT activity for the flooded treatments and with the highest Fe+Pb doses in relation to the control (Fig. 3B). On the other hand, at 6 h AAT, an increase in CAT activity was observed for treatments T5, T8, T9 and T11 and a decrease for treatments T4 and T6 in relation to the control (Fig. 3C). At 12 h AAT, an increase in CAT activity was observed for T2 and T4 and a decrease for treatments T5, T6, T7, T9, T10, T11 and T12 in relation to the control (Fig. 3D). At 24 h AAT, there was an increase in CAT activity for T3, T5, T7, T8, T10 and T12 and a decrease for treatments T2, T4 and T11 in relation to the control (Fig. 3E). At 48 h AAT, there were no significant differences ($p < 0.05$) among treatments, except for treatments T6 and T7 that showed an increase in CAT activity in relation to the control (Fig. 3F).

3.3.4. Guaiacol peroxidase (GPX) activity

Significant differences ($p < 0.05$) were observed for GPX activity in all treatments and leaf sampling times (Fig. 4). There was an increase in

GPX activity at 48 h AAT (Fig. 4F). There was, at 0 h AAT, an increase in GPX activity for T6, T8 and T12 in relation to the control. GPX higher activity was observed for the flooded treatment and with the highest dose of Fe+Pb (T12) (Fig. 4A). However, at 3 h AAT, there were no significant differences ($p < 0.05$) between treatments compared to the control, except for T3, where there was an increase in GPX activity (Fig. 4B). The same fact was observed at 6 h AAT, except for treatments T2 and T9, which showed an increase in GPX activity (Fig. 4C). It was verified, at 12 h AAT, that there was an increase in GPX activity for most treatments, except for T3, T4 and T8, which did not differ significantly ($p < 0.05$) in relation to the control (Fig. 4D). However, at 24 h AAT, an increase in GPX activity was observed for treatments T3, T4, T6 and T8 and a decrease for T2, T7, T9, T10 and T11 in relation to the control (Fig. 4E). On the other hand, at 48 h AAT, a decrease in GPX activity was observed for most treatments compared to the control, except for treatment T12, where an increase in GPX activity was observed, and for treatment T3, which mean value was not significantly different ($p < 0.05$) in relation to the control (Fig. 4F). At 48 h AAT there was a more than 100% increase in treatments for GPX activity compared to 0 h AAT.

3.4. Biplot analysis based on multivariate of principal components analysis

Principal component analysis summarized 64% of the total variance

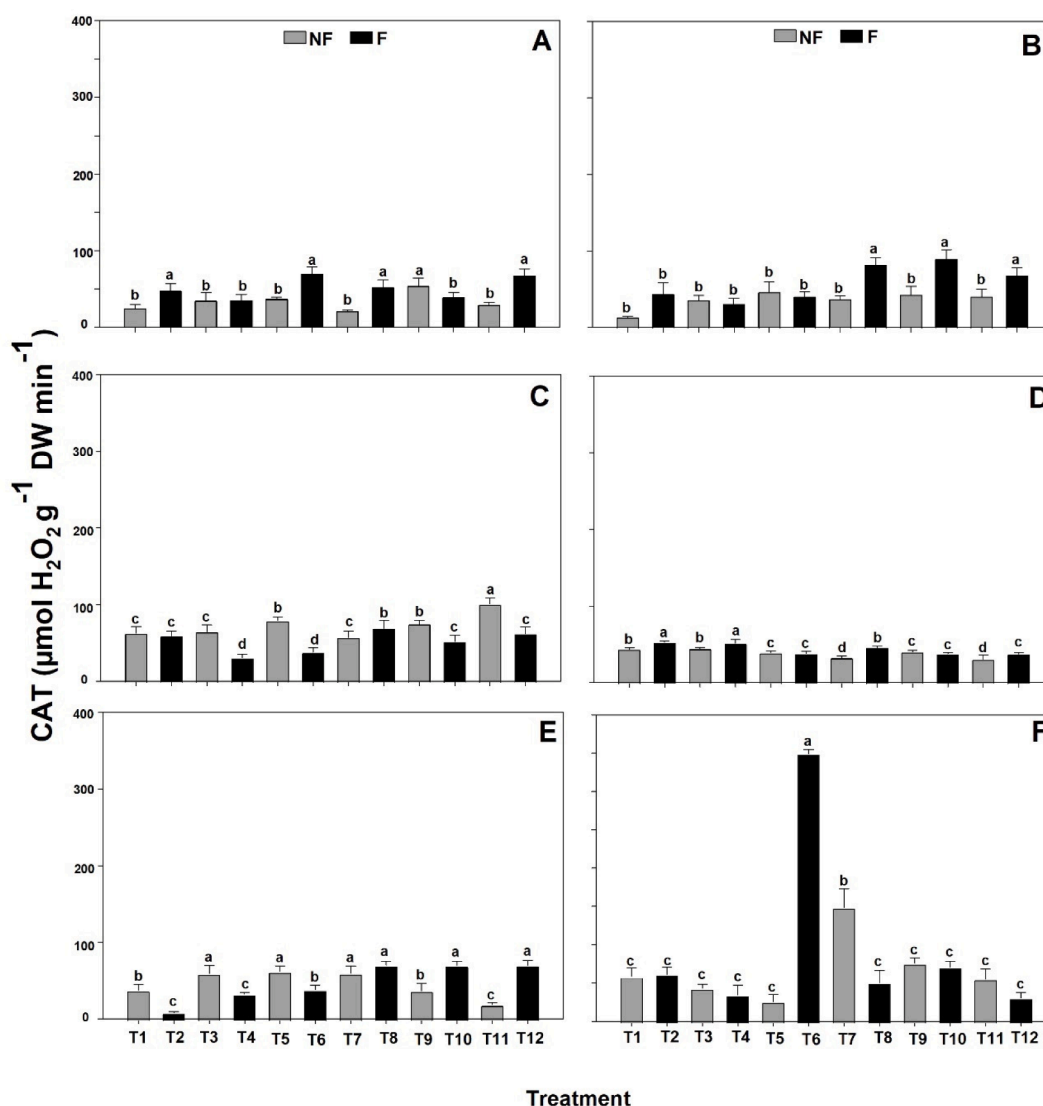


Fig. 3. Catalase (CAT) activity in leaves of juvenile plants of CCN 51 cacao genotype subjected to flooding and doses of Pb and Pb+Fe in the soil at 0 (A), 3 (B), 6 (C), 12 (D), 24 (E) and 48 h (F) after application of treatments. Mean values of four replications ($\pm$ SE). Letters indicate comparisons of means among treatments by the Scott-Knott test ($p < 0.05$). NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

contained in the data matrix, with the percentage of total variance explained by PC1 being 34%, while 30% was explained by PC2. The variable with the highest loading score in PC1 was GPX, while in PC2, the highest loading scores were observed for CAT and APX. Consequently, the percentage of total variance accumulated in PC1 and PC2 was composed, for the most part, by variations originating from GPX, CAT and APX, respectively. According to the angle between the vectors in the plane, there was no correlation between CAT and SOD (Supplementary Fig. 1). The other correlations were positive.

Through the evaluation of the relationship between treatments and variables, it was observed that there was: (i) a positive association of treatments located in the upper right quadrant, with CAT and GPX activities, and a negative association of those located in the lower left quadrant; (ii) positive association of treatments located in the lower right quadrant, with SOD and APX activities, and negative association of treatments located in the upper left quadrant.

According to the dispersion of treatments in the plan, it was verified,

at 0, 3 and 6 h AAT, for all treatments, that there was lower activity of the evaluated enzymes, when compared to the treatments at 24 and 48 h AAT, which showed higher CAT and GPX and SOD and APX activities, respectively (Supplementary Fig. 1). Specifically, at 24 h AAT, the highest SOD and APX activities were observed for treatments T2, T4, T6 and T11, while at 48 h AAT, the highest CAT and GPX activities were observed for T6, T7 and T12.

3.5. Proteomic analysis

3.5.1. Leaf protein profiles in SDS-PAGE

Electrophoretic profiles, in SDS-PAGE, of leaf proteins of juvenile plants of CCN 51 cacao genotype subjected to T1 (control, soil moisture maintained near FC and without addition of Pb or Pb+Fe in soil) and T6 (flooded treatment with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 6 and 24 h AAT, selected from multivariate of principal components analysis, showed well defined bands, well separated and without

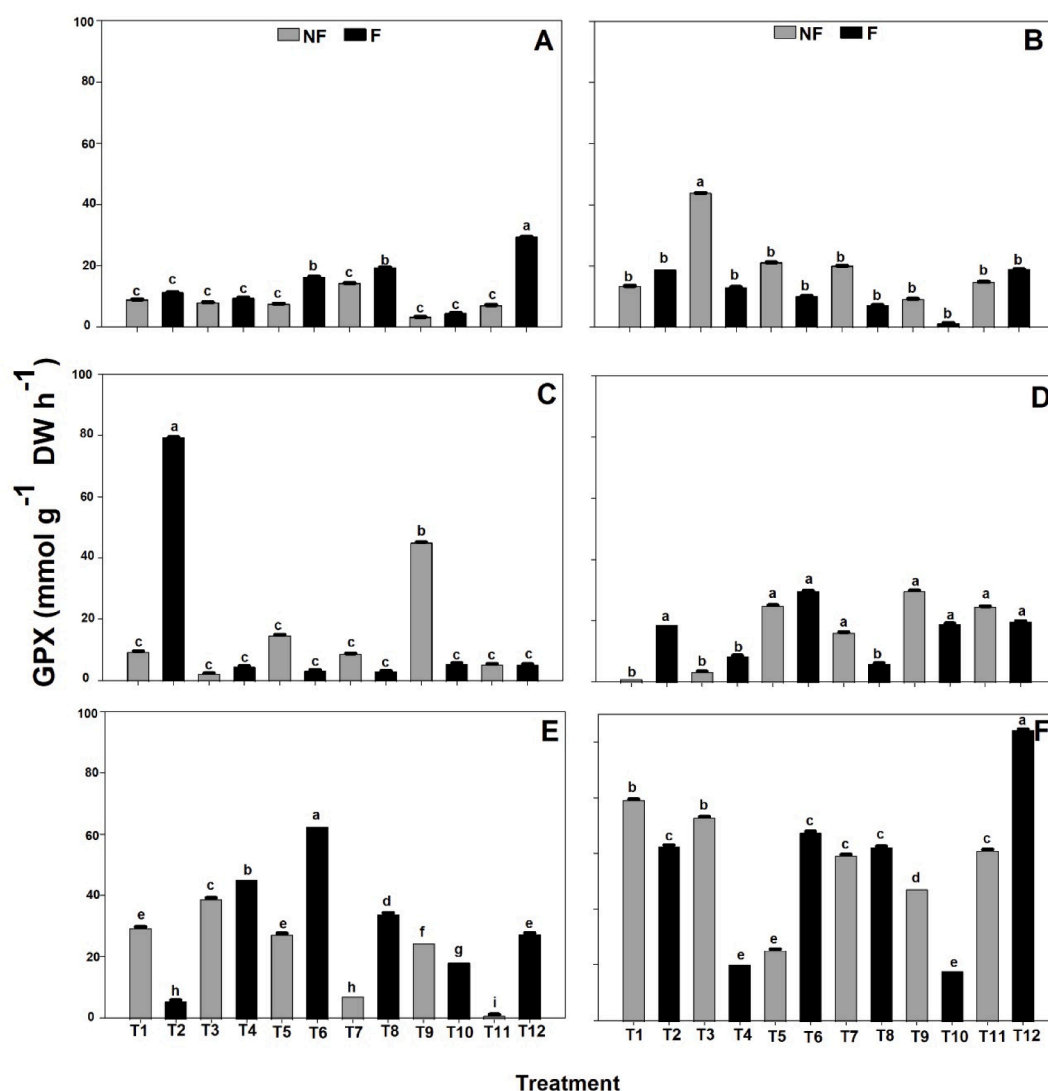


Fig. 4. Guaiacol peroxidase (GPX) activity in leaves of juvenile plants of CCN 51 cacao genotype subjected to flooding and doses of Pb and Pb+Fe in the soil at 0 (A), 3 (B), 6 (C), 12 (D), 24 (E) and 48 h (F) after application of treatments. Mean values of four replications ($\pm$ SE). Letters indicate comparisons of means between treatments by the Scott-Knott test ($p < 0.05$). NF - Not flooding; F - Flooding; T1 - Control treatment - soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil; T2 - Flooded treatment without addition of Pb or Pb+Fe in soil; T3 - NF + 2 mmol Pb kg⁻¹ soil; T4 - F + 2 mmol Pb kg⁻¹ soil; T5 - NF + (2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T6 - F + (Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil); T7 - NF + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T8 - F + (2 mmol Pb kg⁻¹ soil + 1 mmol Fe kg⁻¹ soil); T9 - NF + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T10 - F + (2 mmol Pb kg⁻¹ soil + 1.5 mmol Fe kg⁻¹ soil); T11 - NF + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil) and T12 - F + (2 mmol Pb kg⁻¹ soil + 2 mmol Fe kg⁻¹ soil).

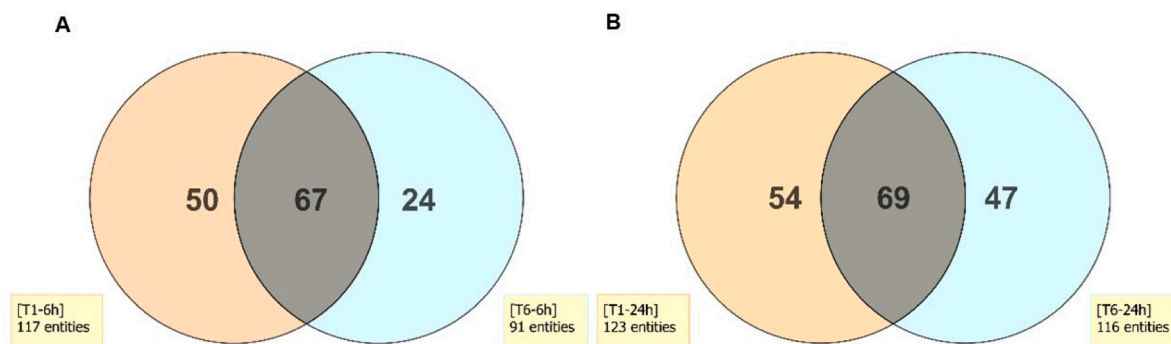


Fig. 5. Venn diagram representing exclusive and common proteins of treatments T1 (control, soil moisture maintained near field capacity and without addition of Pb or Pb+Fe in soil) and T6 (flooded treatment with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil) at 6 h (A) and 24 h (B) after application of the treatments, from leaf samples of juvenile plants of CCN 51 cacao genotype subjected to flooding and Pb+Fe doses in soil.

presentation of drag, thus confirming a good extraction of protein (Supplementary Fig. 2). It was found that the analyzed treatments showed an accumulation of proteins throughout the gel, with more intense bands at molecular masses of 55, 30, 20 and 15 kDa.

3.5.2. Identification of leaf proteins

A total of 1828 proteins were reported in leaf samples whose plants were subjected to flooding and Pb+Fe doses in the soil, collected at 6 and 24 h AAT. Of these proteins, 365 were identified and validated (FDR<1%), of which 203 passed through the quality filter [Score > 5 and SPI (percentage of the extracted spectrum that is explained by the database search result) > 60%]. Of the 365 proteins identified, 50 were unique to T1 at 6 h AAT, 24 were unique to T6 at 6 h AAT, and 67 proteins were common between these two treatments at 6 h AAT (Fig. 5A). In addition to these, of the 365 proteins identified, 54 were exclusive to T1 at 24 h AAT, 47 were exclusive to T6 at 24 h AAT and 69 proteins were common between these two treatments at 24 h AAT (Fig. 5B).

After filtering the identified proteins, by the abundance and frequency criteria, described in the methodology, an unpaired *t*-test analysis was performed between T1 and T6 treatments at 6 and 24 h AAT, to identify the differentially abundant proteins statistically. In addition, proteins were also filtered by fold change. Cutoff was defined as $p < 0.05$ and fold change ≥ 1.5 (Supplementary Fig. 3).

Of the 67 proteins common to the two treatments (T1 and T6) at 6 h AAT, 17 were differentially abundant, 15 exclusives to T1 and two exclusives to T6 (Supplementary Fig. 4). At 24 h AAT, differentially abundant proteins between treatments were not identified. Thus, of the 203 proteins identified, 50 exclusive proteins were from T1 (control) at 6 h AAT (Supplementary Table 2), 24 exclusive proteins were from T6 at 6 h AAT (Supplementary Table 3), 17 differentially abundant proteins were from treatments T1 and T6 at 6 h AAT (Supplementary Table 2), 54 unique proteins were from T1 (control) at 24 h AAT (Supplementary Table 3) and 47 unique proteins were from T6 at 24 h AAT (Supplementary Table 3). For all times plant material samples, a greater number of proteins was identified in the control treatment compared to the treatment with Pb+Fe, thus showing that the doses of metals reduced the number of proteins in juvenile plants of CCN 51. On the other hand, the lowest number of proteins was identified for the treatment with Pb+Fe in soil at 6 h AAT (Supplementary Fig. 5), whereas the highest number was identified in the control treatment at 24 h AAT (Supplementary Fig. 6). It was also observed that there was, at 24 h AAT, a greater number of proteins than at 6 h AAT.

3.5.3. Grouping of differentially abundant proteins, according to higher (Up-regulated) and lower (Down-regulated) abundance

After identifying the proteins, a clustering analysis was performed. Proteins were grouped by their normalized fold change values (Supplementary Fig. 5 and 6). In this way, it was possible to visually identify which proteins had the most contrasting abundances between treatments. The color scale ranges from red, which represents greater protein abundance, through light colors, indicating lower fold change values, to blue, which indicates that the protein was not identified in that treatment.

It could be observed, at 6 h AAT (Supplementary Fig. 5), that the differentially accumulated proteins showed a greater accumulation in treatment T1 (control) than in treatment T6 (Supplementary Fig. 5). Treatment T1 also showed a greater abundance of proteins compared to treatment T6 (Supplementary Fig. 5). At 24 h AAT (Supplementary Fig. 6), there were no differentially accumulated proteins, only exclusive to the treatments. The highest number was shown by T1, but the highest abundance occurred in T6 (Supplementary Fig. 6). The abundance of these proteins was also analyzed according to their involvement in biological processes (Fig. 6).

It was observed, at 6 h AAT, that proteins related to transport and localization had 100% of their abundance decreased. In contrast, proteins related to unknown processes had 35% and 65% of their abundances increased and decreased, respectively. On the other hand, proteins related to stimulus/defense/stress responses had 18% and 82% of their abundances increased and decreased, respectively. Furthermore, proteins related to cellular processes had 42% and 58% of their abundances increased and decreased, respectively. Proteins related to metabolic processes had 23% and 77% of their abundance increased and decreased, respectively, and proteins related to biological regulation had 100% of their abundance decreased (Fig. 6).

It was verified, at 24 h AAT, that the proteins related to the development process had 100% of their abundance decreased. In contrast, proteins related to transport and localization had 50% and 50% of their abundances increased and decreased, respectively. Proteins related to unknown processes had 41% and 59% of their abundances increased and decreased, respectively. Proteins related to stimulus/defense/stress responses had 35% and 65% of their abundances increased and decreased, respectively. Proteins related to cellular processes had 50% and 50% of their abundances increased and decreased, respectively. Furthermore, proteins related to metabolic processes had 62% and 38% of their abundance increased and decreased, respectively, while proteins related to biological regulation had 100% of their abundance decreased

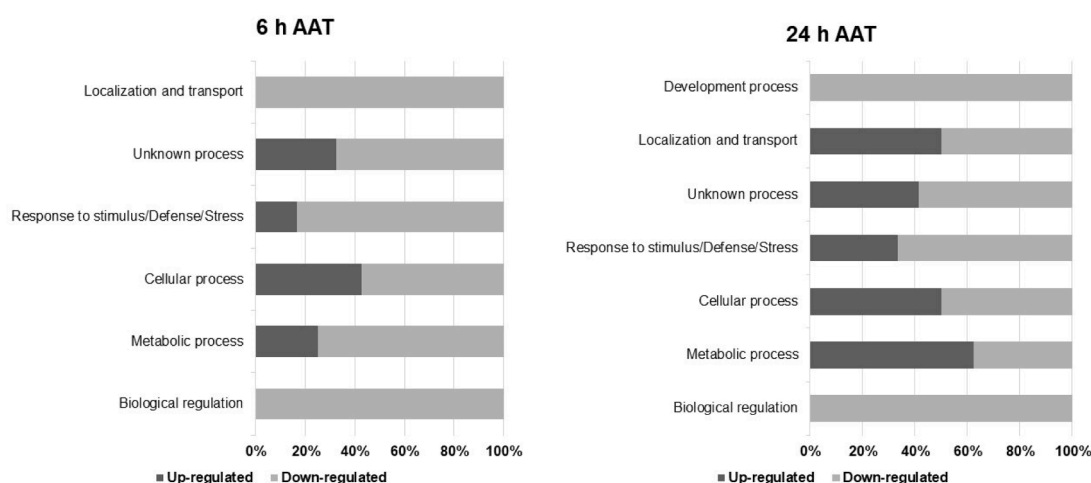


Fig. 6. Distribution of identified proteins, performed according to their involvement in biological processes and expression level, in leaf samples of juvenile plants of CCN 51 cacao genotype subjected to: T1 (control, soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil) and T6 (flooded treatment with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 6 and 24 h after application of the treatments. Bars containing dark gray and light gray colors represent proteins that were increased and decreased in abundance, respectively (ANOVA with $p < 0.05$ and Fold ≥ 1.5).

(Fig. 6). Besides, it was possible to observe that in most of the biological processes, the proteins showed a reduced abundance in the treatment whose plants were subjected to the flooding and to Pb+Fe doses in the soil (T6) in relation to the control (T1), mainly at 6 h AAT (Fig. 6), showing that these metals affected the biological processes of juvenile plants of the cacao genotype CCN 51.

3.5.4. Functional characterization, biological process and cellular component of leaf proteins according to gene ontology

The identified proteins were classified and categorized into groups according to their involvement in metabolic processes (Figs. 7A and 8A), their functions (Figs. 7B and 8B) and cellular component localization (Figs. 7C and 8C), using the analysis of Gene Ontology by *QuickGO* and *PANTHER* software. It could be observed, at 6 h AAT, that 58%, 18%, 16%, 6%, 1% and 1% of the proteins were related to unknown processes, cellular processes, metabolic processes, response to stimulus/defense/stress, biological regulation, and transport and localization, respectively (Fig. 7A). Regarding their function, 69%, 21%, 4%, 3% and 3% of the proteins were related to catalytic activity, binding, molecular function regulator, transporter activity and structural molecule activity, respectively (Fig. 7B). On the other hand, regarding the localization of cellular component, 59% and 41% of the proteins were localized in the cellular anatomical entity and in the protein-containing complex (Fig. 7C).

It was verified, at 24 h AAT, that 63%, 12%, 16%, 3%, 1%, 4% and 1% of the proteins were related to unknown processes, cellular processes, metabolic processes, the response to stimulus/defense/stress, biological regulation, transport and localization, and the developmental process, respectively (Fig. 8A). Regarding their function, 56%, 25%, 2%, 6%, 4%, 5% and 2% of the proteins were related to catalytic activity, binding, molecular function regulator, transporter activity, structural molecule activity, ATP-dependent activity, and translation regulator activity, respectively (Fig. 8B). In contrast, regarding the cellular component of localization, 92% and 8% of the proteins were located in the cellular anatomical entity and in the protein-containing complex,

respectively (Fig. 8C).

4. Discussion

In the present study, the doses of Pb or Pb+Fe applied to the soil did not significantly interfere with the predawn leaf water potential (Ψ_{WL}), whereas soil flooding promoted a decline of Ψ_{WL} over the exposure time of juvenile plants of the cacao genotype CCN 51 to this stressor (Table 1). Da Silva Branco et al. (2017), when evaluating cacao genotypes contrasting for tolerance to flooding, grown at different light intensities, also found that flooding was the factor that had the greatest influence on most physiological responses. Soil flooding directly interferes with the diffusion of oxygen in plant tissues, the diffusion of gas between cells and limits oxygen exchange and mitochondrial respiration, seriously affecting the normal physiological and biochemical processes of the plant (Xu et al., 2006; Liu et al., 2012; Voesenek and Baileyserres, 2013). In conditions of flooding, there is decrease in the uptake of water and soil mineral nutrients, via the root system, mainly due to the decrease in root conductance, resulting in nutritional deficiency and compromising the growth and development of the plant (Steffens et al., 2005).

Juvenile plants of CCN 51 accumulated more Fe and Pb in the roots than in the leaves, only a small fraction was translocated to the shoot, regardless of the water content in the soil (Table 2), a fact also observed in several other plant species (Feng et al., 2013). Normally, uptaken Pb is accumulated in the roots and only a small fraction is transported to the plant shoot (Kumar et al., 2017). Flooding and the variation of Pb+Fe doses in the soil did not interfere with the transport of Pb to the shoot of the plants since there were no major alterations in the levels of Pb in the leaves. The transport of Pb from soil to plants and its distribution in plant parts depends on several external and internal factors, which can act individually and, or in interaction with each other (Punamiya et al., 2010; Ashraf et al., 2015). Mobility, solubility, and bioavailability of Pb in soil largely affect its uptake by plant roots (Punamiya et al., 2010; Ashraf et al., 2015). Root Pb uptake occurs through divalent cation

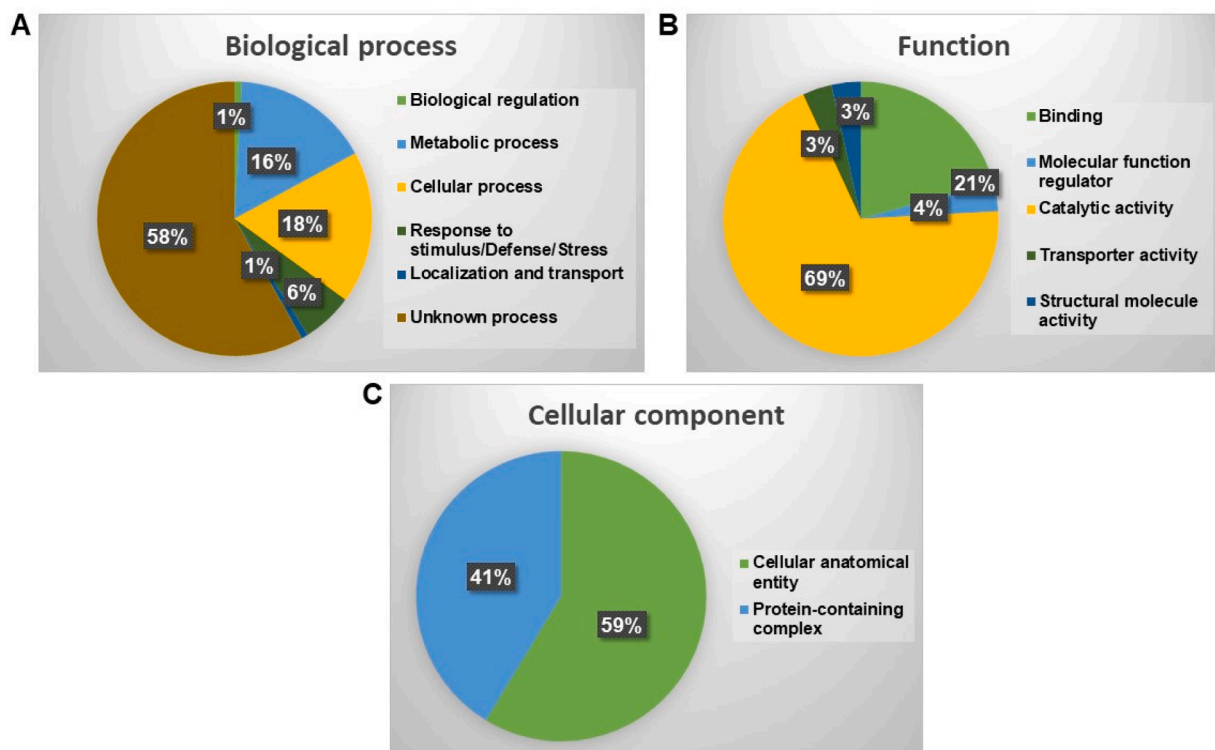


Fig. 7. Functional characterization, biological process and cellular component of leaf proteins, according to gene ontology, in leaf samples of juvenile plants of CCN 51 cacao genotype subjected to: T1 (control, soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil) and T6 (flooded treatment with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 6 h after application of the treatments. A - Biological process; B - Function; C - Cellular component.

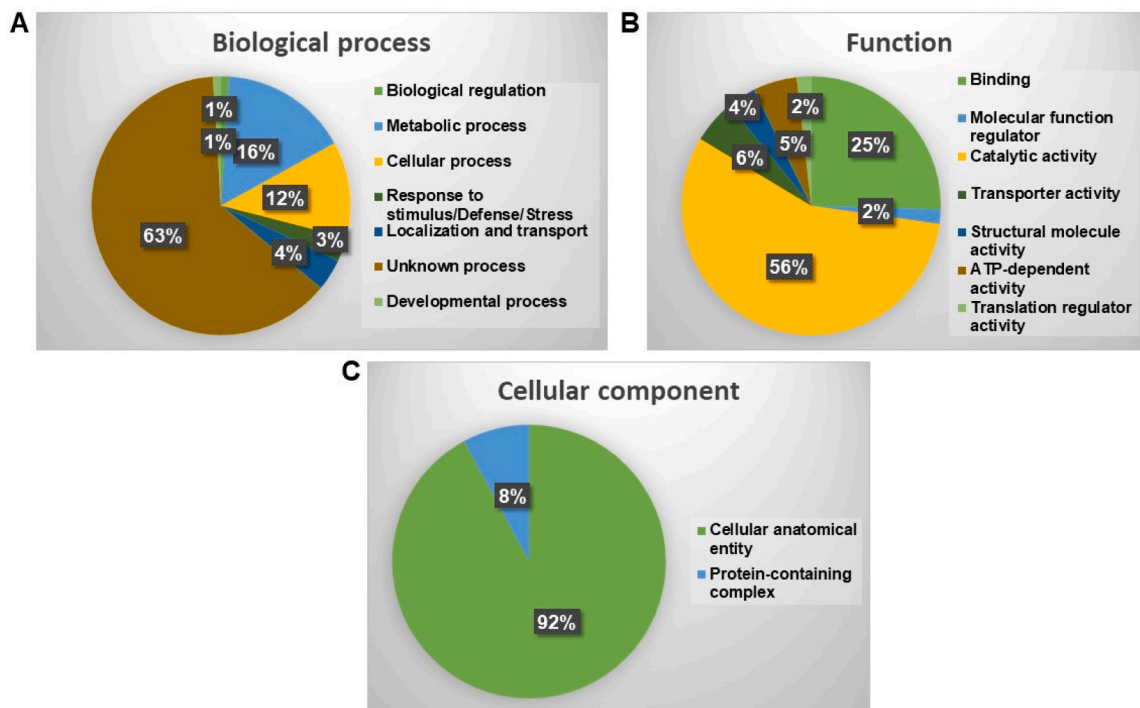


Fig. 8. Functional characterization, biological process and cellular component of leaf proteins, according to gene ontology, in leaf samples of juvenile plants of CCN 51 cacao genotype subjected to: T1 (control, soil moisture maintained near field capacity and without addition of Pb or Pb+Fe to the soil) and T6 (flooded treatment with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 24 h after application of the treatments. A - Biological process; B - Function; C - Cellular component.

transporters found in the plasma membrane, such as ZIP, Zn and Fe transporters (Auguy et al., 2013). In the present study, there was a greater uptake and accumulation of Fe in the plant roots in relation to Pb (Table 2).

The complex defensive antioxidant system in plants can protect cell membranes and organelles from the harmful oxidative effects of ROS (Foyer et al., 1991). Key antioxidant enzymes such as superoxide dismutase (SOD), guaiacol peroxidase (GPX), ascorbate peroxidase (APX), catalase (CAT) and glutathione reductase can effectively minimize ROS activity (Foyer et al., 1991). In the present work, it was possible to observe that there was an increase in SOD activity, at 12 h AAT, in the treatments with flooded soil (Fig. 1). The SOD enzyme is recognized as the first line of cellular defense, catalyzing the dismutation of superoxide anion into H₂O₂ and O₂ in different cellular compartments (Gill and Tuteja, 2010; Rajput et al., 2021). Yadav et al. (2019) showed that the expression profile of SOD genes is upregulated under different abiotic stresses. This demonstrates that during this flooding period there was oxidative stress in the flooded plants and, consequently, an increase in SOD activity in relation to the control. On the other hand, regarding the doses of Pb or Pb+Fe applied in the soil, not many differences were observed between treatments, demonstrating again that the doses of metals were not a limiting factor in the SOD activity, but the flooding of the soil.

The highest APX, CAT and GPX activities were observed at 48 h AAT, with the exception of APX, whose activity also increased at 24 h AAT (Figs. 2, 3 and 4). This demonstrates that, during a longer period of exposure to Pb or Pb+Fe doses and soil flooding, there was an intensification of oxidative stress in juvenile plants of CCN 51. APX, CAT and GPX have the potential to convert H₂O₂ into H₂O and O₂, being important for ROS detoxification in plants subjected to stressful conditions (Montillet et al., 2005; Gill and Tuteja, 2010). The increase in antioxidative activity enhances the plant's ability to detoxify free radicals and prevents the metal-induced oxidative burst in the plant cell (Verma et al., 2013). An increase in CAT activity is considered direct evidence of oxidative damage (Sairam and Sexena, 2000; Khan et al.,

2007). Apraez Muñoz et al. (2021) and Reis et al. (2015) also observed high SOD, APX, CAT and GPX activities in juvenile cacao plants exposed to the toxic effects of Pb. Furthermore, studies have shown that the increase in Fe content, parallel to the increase in Pb content in the roots, may indicate that Fe is promoting ROS detoxification, as in roots of *Chloris gayana* (Matés, 2000) and in *Vallisneria natans* (Wang et al., 2011).

As the states of cell function are determined by the differential expression of cellular proteins (de Sousa Abreu et al., 2009), proteomic analysis can reveal relationships between protein abundance and plant acclimatization to Pb stress and provide valuable candidate gene resources for phytoremediation (Shen et al., 2021). In the present work, the leaf protein profiles of juvenile plants of the CCN 51 cacao genotype, subjected to T1 (control, soil moisture maintained near FC and without addition of Pb or Pb+Fe to the soil) and T6 (flooded soil with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 6 and 24 h AAT, selected from multivariate analysis of principal components, demonstrated that most of the proteins was included in the categories of (i) cellular process, (ii) metabolic, (iii) stimulus/defense/stress response, (iv) localization and transport, (v) biological regulation, (vi) developmental process, and (vii) unknown. At both treatments, at 6 h and 24 h AAT, most proteins were categorized into unknown processes. Furthermore, within each category, most proteins were repressed, particularly at 6 h AAT (Fig. 6).

In the present study, leaf proteomic analysis, performed on juvenile plants of CCN 51, subjected to T1 and T6 treatments, at 6 h and 24 h AAT, demonstrated that Pb+Fe and soil flooding induced a decline in soluble protein contents (Supplementary Fig. 5 and 6). The reduction in protein contents may be due to degradation by proteases as a result of high metal exposure (Romero-Puertas et al., 2002; Hou et al., 2007). Similar studies performed by Mohan and Hosetti (1997) and Singh et al. (2010) demonstrated that there is a decrease in plant protein levels induced by the accumulation of Pb in plant tissues.

Within the cellular process category, four leaf proteins were decreased and three were increased in abundance in T6, at 6 h AAT. Among the four proteins, reduced in abundance, is ribulose-1,5-

bisphosphate carboxylase/oxygenase (Rubisco) (Line 1, Supplementary Table 2), a key enzyme of the Calvin & Benson cycle and the most abundant in plants, compartmentalized in chloroplasts (Siedlecka et al., 1998). Generally, abiotic stresses negatively affect photosynthesis at different levels, such as CO₂ diffusion, PSII efficiency, electron transport, ROS formation, Rubisco content (dependent on ATP and NADPH supply), Rubisco activity (Qureshi et al., 2010; Bashir et al., 2015) and photorespiration (Saibo et al., 2009). In the present study, in conditions of soil flooded, containing 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil, Fe failed to mitigate the toxic effect of Pb on the Rubisco in leaf of juvenile plants of CCN 51. Another protein also decreased in abundance was histone H2B (Line 2, Supplementary Table 2). Histones are the main proteins that make up the nucleosome, rich in lysine and arginine residues, and, through electrostatic interactions, neutralize the DNA, allowing it to be strongly compacted (Luger et al., 1997). These proteins play a significant role in gene regulation, are the main protein components of chromatin and can undergo post-translational modifications, which modify the structural/functional properties of chromatin (Houben et al., 2007). Chromatin organization can be altered by the incorporation of histone variants, and this mechanism, different from deposition, makes them adaptable to respond to environmental stimuli (Talbert and Henikoff, 2014). In addition to heavy metals, hypoxia and oxidative stress also induce epigenetic dysregulation, which results in changes in gene expression, probably responsible for its toxic effects (Costa et al., 2011). In addition, the enzymes that modify histones can also be affected by exposure to the metal (Costa et al., 2011).

In the T6 treatment, at 24 h AAT, six proteins involved in the cellular process were decreased and six were increased in abundance. Among the proteins reduced in abundance, we have glutathione transferase (GST) (Line 3, Supplementary Table 3), which in addition to acting in antioxidant metabolism, in the removal of ROS, this enzyme participates in the glutathione-ascorbate cycle (ASA-GSH) (Anjum et al., 2012). In this process, ascorbate is used as a substrate in the synthesis of glutathione, which is composed of a group of multifunctional enzymes, and catalyzes the conjugation of glutathione (GSH) with heavy metals, which are stored in the vacuole for cellular detoxification in plants (Freeman et al., 2004; Gupta et al., 2010). Besides that, in *Macrotyloma uniflorum* and *Cicer arietinum*, subjected to different doses of Pb, there was an increase in the activity of GST to combat oxidative stress (Van Assche and Clijsters, 1990). On the other hand, among the proteins increased in abundance, was Chaperonin 20 (Line 12, Supplementary Table 3). Molecular chaperonin plays a vital role in the stabilization, folding, assembly and translocation of proteins under abiotic stress (Komatsu and Hossain, 2013). In the work by Cheng et al. (2018), the abundance of chaperonin 20 (CPN20) was increased under osmotic and Cd stress after four days of treatment. CPN20 may mediate Fe-SOD activation for protection against oxidative stress and in chloroplast development (Sarry et al., 2006; Kuo et al., 2013). Thus, we can infer that Fe, applied together with Pb in flooded soil, contributed to the activation of the Fe-SOD enzyme, and, consequently, to the protection against oxidative stress caused by Pb in juvenile plants of CCN 51.

Within the metabolic process category, six proteins had their abundance decreased and two had their abundance increased in T6, at 6 h AAT. Among the proteins that had their abundance decreased, we have glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (Line 8, Supplementary Table 2). GAPDH plays a key role in glycolysis, gene transcription, DNA replication, DNA repair, and nuclear RNA export (Singh and Green, 1993). Vescovi et al. (2013) found that the cytosolic GAPDH isoenzyme GAPC1 may play a role in oxidative stress signaling or protection in *Arabidopsis* roots under stress. On the other hand, the proteins that had their abundance increased were two fructose bisphosphate aldolases (Line 9 and 15, Supplementary Table 2). This enzyme is a constituent of the glycolytic/gluconeogenic pathway and the pentose phosphate cycle in plants. Although aldolase catalyzes a readily reversible reaction, it has no known regulatory properties and appears irrelevant to the control of metabolism and growth. Small changes in

their activity have marked consequences for photosynthesis, carbon partitioning and growth (Haake et al., 1998).

In treatment T6, at 24 h AAT, still within the metabolic process category, six proteins had their abundance decreased and 10 proteins had their abundance increased. Among the proteins reduced in abundance, we have, again, the enzyme fructose-bisphosphate aldolase (Lines 15 and 16, Supplementary Table 3), showing that the increase in the exposure time of juvenile plants of CCN 51 to the treatment with flooded soil containing Pb+Fe significantly affected the glycolytic/gluconeogenic pathway and the pentose phosphate cycle. On the other hand, among the proteins that had their abundance increased we have triose phosphate isomerase (TPI) (Line 23, Supplementary Table 3). TPI catalyzes the interconversion of glyceraldehyde-3-phosphate into dihydroxyacetone phosphate, metabolites that are part of different routes of carbohydrate metabolism for energy generation and formation of carbon skeleton for biosynthetic reactions (Curto et al., 2006). TPI overexpression is necessary when carbon resource is needed to synthesize amino acids and proteins (Dorion et al., 2005).

Within the stimulus/defense/stress response category, for T6 at 6 h AAT, five proteins had their abundance decreased and one increased. Among the proteins that had their abundance decreased, all are related to reactions involved in photosynthesis, such as the protein binding to chlorophylls *a* and *b* (Lines 16, 17, 18 and 20, Supplementary Table 2). This chloroplast protein makes up the light harvesting complex of PSII of the photochemical phase of photosynthesis, which favors light uptake and the transfer of excitation energy to the PSII reaction center during charge separation (Takahashi et al., 2006). In a study with *Spinacia oleracea*, the contents of chlorophylls *a* and *b*, as well as their proportions, strongly decreased with increasing Pb dose (Xiao et al., 2008). On the other hand, the protein that had its abundance increased was alcohol dehydrogenase (ADH) (Line 21, Supplementary Table 2). ADH is a Zn-binding enzyme that depends on NAD(P) cofactors to participate in the conversion between ethanol and acetaldehyde (Höög et al., 2003; Thompson et al., 2007). One of the main effects of soil flooding is O₂ deprivation in the root zone (Folzer et al., 2005). In the privation (anoxia) and, or under conditions of low availability of O₂ in the soil (hypoxia), glycolysis and fermentation can exceed aerobic metabolism and become the only route of energy production, leading to the accumulation of various fermentation products, such as ethanol (highly toxic), lactate and alanine, derivatives of pyruvate (Taiz and Zeiger, 2017). As ADH activity is considered a necessary condition for plant survival under anaerobic conditions, the response of the plant ADH gene to anaerobic stress has always been the focus of research (Liskens, 1966; Harberd, 1982; Bailey-Serres and Voisenek, 2008). Bertolde et al. (2014), when working with juvenile plants of two contrasting *T. cacao* genotypes for tolerance to flooding, found that there was a significant increase in ADH activity in the tolerant genotype, compared to the intolerant genotype, when the plants were subjected to flooding for 24 h.

In treatment T6, at 24 h AAT, two proteins had their abundance increased and one had its abundance decreased. Unlike T6, at 6 h AAT, one of the proteins that had its abundance increased was chlorophyll binding protein *a-b* (Line 31, Supplementary Table 2), whereas the protein that had its abundance decreased was ADH (Line 29, Supplementary Table 3). This may have occurred due to the longer exposure time of juvenile cacao plants to Pb+Fe doses and soil flooding, which favored Fe uptake in Pb detriment and chlorophyll *a-b* binding protein biosynthesis. On the other hand, the decrease in ADH abundance is largely due to Pb toxicity, due to the fact that the CCN 51 is tolerant to flooding, as observed by Bertolde et al. (2010) when evaluating the tolerance of 35 clonal cacao genotypes to flooding for 45 days. Transcriptome and proteome analyses have identified numerous genes that are expressed in response to stress induced by O₂ deficiency in plants (Tadege et al., 1998; Klok et al., 2002; Huang et al., 2005; Liu et al., 2005; Alam et al., 2010), providing evidence of the complexity of plant responses to this stressor factor. In addition to O₂ deficiency induced stress, the plant proteome response to Pb stress is a diverse process that

depends on chemical forms, stress intensity, stress duration, and plant species (Lyu et al., 2020).

In the present study, from the hierarchical grouping, based on protein accumulation profiles involved in responses to stress induced by Pb+Fe and soil flooding, it was possible to verify that, among the 50 proteins verified exclusively in treatment T1 (control) at 6 h AAT (Fig. 5A), the most abundant (Supplementary Fig. 5) was entered into the unknown process category [PSI subunit V (Line 53, Supplementary Table 2)]. Furthermore, among the 24 proteins found exclusively for the T6 treatment, at 6 h AAT (Fig. 5A), the most abundant (Supplementary Fig. 5) is also included in this same category [ATP synthase subunit beta (Line 73, Supplementary Table 2)]. Furthermore, the most abundant protein between T1 and T6 is also included in this same category [Ribulose biphosphate carboxylase small chain (Line 25, Supplementary Table 2)].

PSI is one of the four main complexes responsible for the reactions of the photochemical phase of photosynthesis, bound to the thylakoid membranes of chloroplasts, which uses light to catalyze plastocyanin oxidation and NADP⁺ reduction, in addition to contributing to the transmembrane pH gradient (Taiz and Zeiger, 2017). The PSI complex consists of at least 18 different types of protein subunits, 14 of which form the so-called core complex (Taiz and Zeiger, 2017). As the PSI subunit V protein, found in the present study, was the most abundant in T1 (control), this fact corroborates the explanation that Pb was the predominant factor that interfered in the leaf gas exchange of juvenile plants of CCN 51, since it is a flooding tolerant genotype. Araújo et al. (2017), studying Pb toxicity in young plants of the CCN 51, observed that leaf gas exchange was affected with doses of 0.5 and 1.0 g Pb kg⁻¹ soil.

Mitochondrial ATP synthase is the main enzyme complex of energy metabolism, supplying ATP to the cell. Subunits of this enzyme are overexpressed under stress conditions, providing additional energy necessary for maintaining cellular homeostasis (Moghadam et al., 2012). This enzyme is responsible for several regulatory mechanisms, modulating its activity according to environmental changes (Konno et al., 2006). The β -ATP synthase subunit participates in the transport of protons across the membrane, generating an electrochemical gradient, and in the regulation of ATP synthesis (Andersson et al., 2009). Thus, we can infer that soil flooding and Pb+Fe doses applied to the soil induced an increase in the abundance of this protein, aiming to mitigate the stress caused by these stressor factors in juvenile plants of the CCN 51. Overexpression of the ATPase-type protein under Pb stress suggests increased active transport across membranes to maintain cellular homeostasis under toxic metal stress (Baig et al., 2018). On the other hand, Rubisco, which catalyzes the assimilation of CO₂ by the carboxylation of ribulose-1,5-bisphosphate, during the photosynthetic assimilation of carbon (Ellis, 1979), is responsible for up to 30–70% of the soluble leaf proteins (Bah et al., 2010). Therefore, its abundance was higher in both treatments T1 and T6, at 6 h AAT, mainly in T1.

In the present study, from the functional categorization of proteins (Figs. 7 and 8), we were able to verify that in the T6 treatment, both at 6 h and at 24 h AAT, most of the proteins found in juvenile plants of CCN 51, subjected to the Pb+Fe doses and to soil flooding, is involved with (i) catalytic activities (69% at 6 h and 56% at 24 h AAT) and binding (21% at 6 h and 25% at 24 h AAT) (Figs. 7B and 8B), (ii) cellular localization, mainly in the anatomy of the cell (59% at 6 h and 92% at 24 h AAT) and (iii) protein complex (41% at 6 h and 8% at 24 h AAT) (Figs. 7C and 8C). Wang et al. (2016) also observed the same fact when evaluating proteins from roots of *Raphanus sativus* exposed to Pb. Shen et al. (2021), in proteomic studies with the genus *Populus* and Pb toxicity, found that most proteins belonged to the “protein” and “RNA” categories, suggesting that these processes were greatly affected during Pb stress. Furthermore, Baig et al. (2018) observed, in the cellular component characterization, that there was decrease in membrane and organelle activity, while in the molecular function characterization, there was increase in oxidoreductase activity and binding to proteins with

decrease in the binding of calcium and metal ions in plants of *Glycine max* under Pb stress. Thus, plants respond to the toxic effects of metals by altering the expression of genes and proteins associated with stress, leading to the activation of signaling networks for stress regulation (Thapa et al., 2012). Protein modification and the balance between synthesis and degradation are critical forms of regulation, which are coordinated to achieve a unified cell under abiotic stress (Hinkson and Elias, 2011). Thus, in general, it was possible to verify that Pb+Fe doses and soil flooding affected the global proteome of juvenile plants of the CCN 51 cacao genotype, in order to condition a protein repression.

5. Conclusions

Doses of Pb or Pb+Fe did not interfere with the predawn leaf water potential (Ψ_{WL}) of juvenile plants of the CCN 51 cacao genotype, but soil flooding was a key factor for the decline of the values of Ψ_{WL} . In juvenile plants of CCN 51, there was a differential accumulation of Pb and Fe in roots and leaves, both in flooded and non-flooded soil. Most of Fe and Pb uptake by the root system predominantly accumulated in the roots and only a small fraction was translocated to the shoot, regardless of the water content in the soil. Under flooded and non-flooded soil conditions, the variation in Pb+Fe doses in the soil did not interfere with the transport of Pb to the shoot of juvenile plants of CCN 51 since there were no major alterations in the Pb contents in leaves.

Leaf enzymatic antioxidant metabolism of juvenile plants of the CCN 51 was altered, mainly by soil flooding, in relation to the doses of Pb or Pb+Fe combined. The time of exposure to stressor factors (flooded soil and doses of Pb or Pb+Fe) was important for the changes in the leaf antioxidant metabolism of juvenile plants of CCN 51, since there was an increase in the activity of the enzymes CAT, APX and GPX, especially at 48 h after application of the treatments. T6 treatment (flooded soil with 2 mmol Pb kg⁻¹ soil + 0.5 mmol Fe kg⁻¹ soil), at 6 and 24 h after treatment application, had a greater influence on Ψ_{WL} , antioxidant metabolism and leaf proteome of juvenile plants of CCN 51. Doses of Pb+Fe and soil flooding affected leaf proteins of juvenile plants of CCN 51 related to photosynthesis, DNA, antioxidative system and carbohydrate metabolism, whose protein abundance was overexpressed or repressed. Leaf proteins, with greater abundance, from juvenile plants of CCN 51 subjected to the T6 treatment, belong to unknown processes and are related to photosynthesis, energy metabolism and protease inhibition. In general, most identified leaf proteins, from juvenile plants of CCN 51, subjected to the T6 treatment, are related to catalytic functions and binding, with cellular localization and in protein complex.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability

Not applicable.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

CRedit author statement

All authors contributed equally to this work.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.scienta.2023.112420](https://doi.org/10.1016/j.scienta.2023.112420).

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